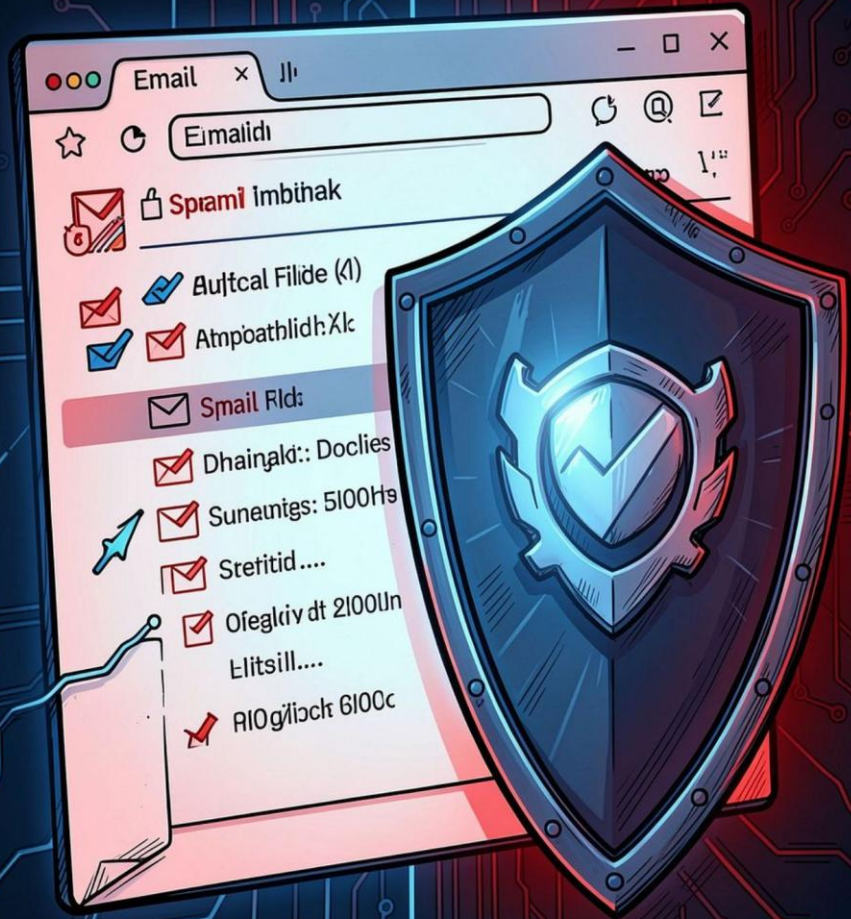




Spam Email Classifier

Machine Learning-Based Email Classification for Customer Service

B.TECH CSE INTERNSHIP REPORT · IILM UNIVERSITY, GREATER NOIDA

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Declaration & Acknowledgement

Candidate's Declaration

Amitesh Kumar declares this report original, not submitted elsewhere, with all references quoted.

Acknowledgement

Gratitude expressed to mentors, Codec Technologies, and family for guidance and support throughout the internship.



Internship Offer & Certificate



Codec Technologies – AI Intern

1-month hybrid internship (10/06/2026 – 10/07/2026), Pan India. Global IT consultancy operating across **27+ countries**. Reporting to assigned Project Head(s); exceptional interns eligible for scholarship awards.

Designation

Artificial Intelligence Intern

Mode

Hybrid, Pan India

Certified

ISO 9001:2015



Introduction & Problem Statement

Email inboxes mix legitimate requests with spam, phishing, and ads. Manual filtering is **inefficient at scale**, consuming time and creating security risks.

The Core Problem

Build a classifier distinguishing spam from legitimate customer emails while controlling false classifications.

Key Challenges

Noisy text, class imbalance, high-dimensional vocabulary, overfitting, and balancing false positives vs. negatives.

Business Impact

Misclassified customer complaints carry significant business cost — precision, recall, and F1-score matter beyond accuracy.

Project Objectives

01

Dataset

Collect labeled spam/ham emails; remove duplicates and empty records.

03

Engineer Features

Create n-grams, TF-IDF weights, message length, URL count, capitalization ratio.

05

Improve & Ensemble

Apply hyperparameter tuning and ensemble methods for robustness.

02

Preprocess

Normalize text via tokenization, lowercasing, stop-word handling.

04

Train & Compare

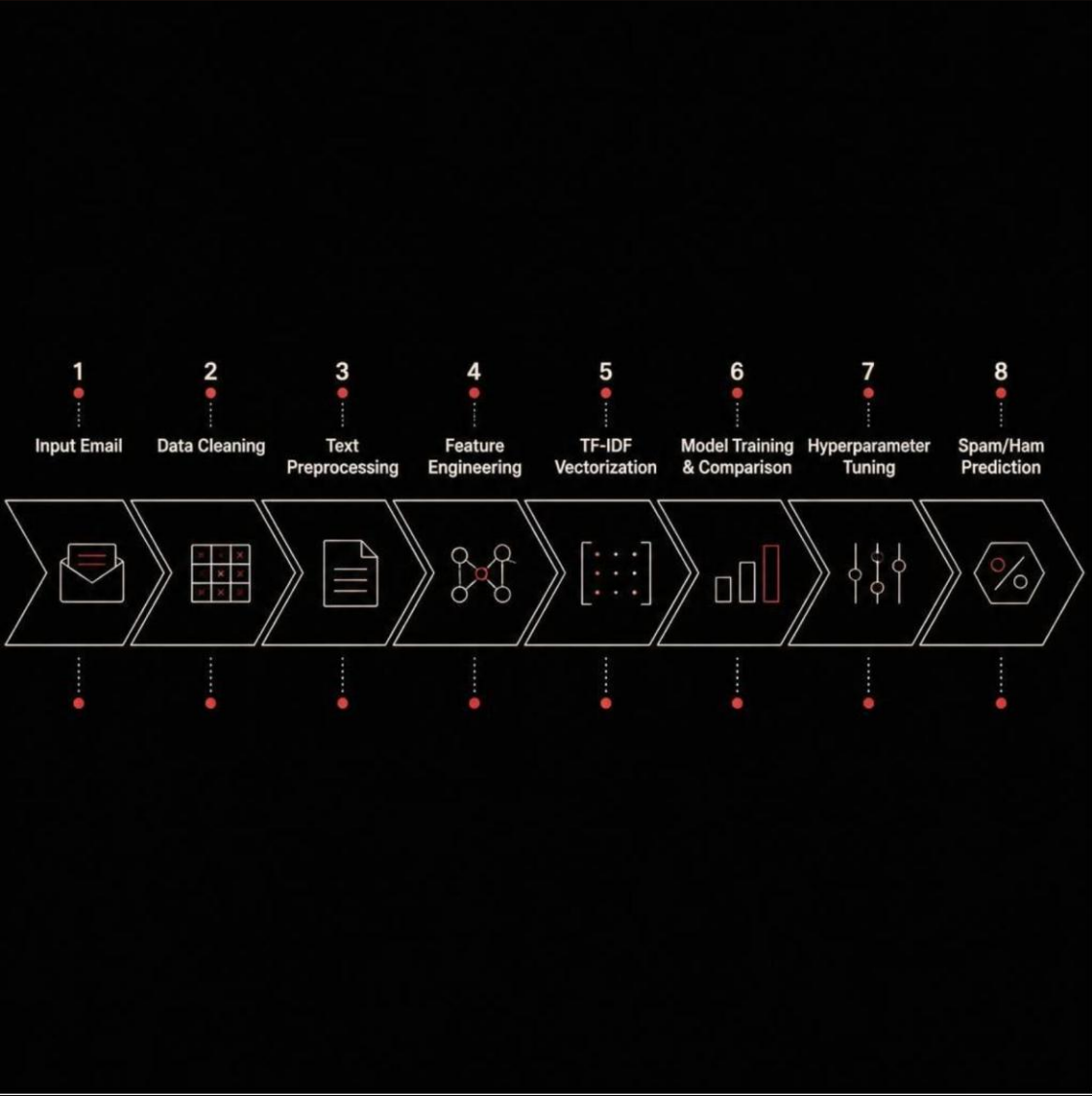
Train Naive Bayes, Logistic Regression, SVM, and tree-based models.

06

Evaluate

Measure accuracy, precision, recall, F1-score, and confusion matrix.

Technologies & System Architecture



Technology Stack

Python Core programming language	Pandas & NumPy Data manipulation
Scikit-learn Vectorization, classification, tuning	NLTK / spaCy NLP preprocessing
Matplotlib / Seaborn Visualization	Jupyter & Git Experimentation & version control

Data Cleaning & Feature Engineering

Data Cleaning Operations

Issue	Action
Missing subject/body	Placeholder or remove record
Duplicate emails	Detect and remove
HTML tags	Strip markup
URLs / addresses	Normalize with tokens
Extra whitespace	Collapse repeated spaces
Labels	Standardize to spam/ham

Engineered Features

→ Word & Character n-grams

Capture phrases and sub-word patterns

→ TF-IDF Weights

Term importance relative to corpus

→ Metadata Features

Message length, URL count, special-character frequency, capitalization ratio

Methodology: Models & Evaluation

Naive Bayes

Fast, effective for TF-IDF. Assumes conditional independence.

Logistic Regression

Strong linear baseline with interpretable feature weights.

Linear SVM

Effective in high-dimensional sparse spaces.

Random Forest

Useful with mixed engineered metadata features.

Ensemble

Voting/stacking combines complementary models for robustness.

80/20 train-test split with stratification. TF-IDF fitted only on training data. Evaluation: accuracy, precision, recall, F1-score, confusion matrix.

Model Improvement & Expected Outcomes

Improvement Strategy



Error Analysis

Inspect misclassified messages for patterns (promotional language, HTML-heavy content).



Hyperparameter Tuning

Adjust regularization, n-gram ranges, vocabulary size, and class weights.



Threshold Tuning

Balance recall vs. precision per business needs.

Expected Outcomes

Automated Detection

Reduces manual inbox screening

Prioritization

Agents focus on genuine requests

Measurable Performance

Precision, recall, F1, confusion matrix

Extensible Architecture

Supports future routing and priority models

Conclusion & Future Scope

This project demonstrates how **NLP and supervised ML** solve a real business problem — combining data cleaning, TF-IDF vectorization, model comparison, and ensemble learning in one pipeline.

Key Takeaway

Precision, recall, F1-score, and error analysis are essential — accuracy alone is insufficient for customer-service impact.

Future Scope

Multilingual support, phishing detection, real-time integration, confidence-based human review, and feedback-driven retraining.

- With appropriate privacy and security controls, this classifier can become a useful component of a broader customer-service automation platform.

